ASE 230 Tools Installation - MySQL

Install on macOS

```
brew install mysql
brew services start mysql
```

Install on Ubuntu / WSL2

```
sudo apt update
sudo apt install -y mysql-server
sudo service mysql start
```

After installation, start the database server with `sudo service mysql start`.

RECOMMENDED

Recommended command for Ubuntu / WSL2

On a regular Ubuntu installation, both `sudo service mysql start` and `sudo systemctl start mysql` normally work.

- `systemctl` is Ubuntu's standard command when `systemd` is running.

RECOMMENDED

For this course, always use `sudo service mysql start` on both Ubuntu and WSL2; You do not need to check whether `systemd` is enabled.

- Many WSL2 installations do not run `systemd` (the Linux service manager), so `systemctl` reports an error instead of starting MySQL.
- The `service` command works on standard Ubuntu and whether or not WSL2 has `systemd` enabled.

Verify on Ubuntu / WSL2

```
mysql --version
sudo service mysql start
sudo mysql -e "SELECT VERSION();"
```

On many Ubuntu installations, MySQL uses `auth_socket` authentication for `root`. Use `sudo mysql` rather than `mysql -u root -p`.

Verify on macOS

```
brew services start mysql
mysql --version
mysql -u root -e "SELECT VERSION();"
```

Homebrew normally configures MySQL without a root password. If you set one, use `mysql -u root -p` instead.

MySQL as a Service

A **service** runs in the background, separate from your terminal. MySQL keeps listening for connections until you stop it.

- After it starts, MySQL continues running even when you close the terminal.
- Commands such as `mysql` and PHP programs connect to that running service; they do not start a new MySQL server.

Before running SQL commands or a PHP program that uses MySQL, the MySQL service must be running.

- If it is stopped, connection attempts will fail even when PHP and MySQL are installed correctly.
- Use the commands on the next slides to start MySQL, check its status, stop it only when needed, or restart it after changing its configuration.

MySQL Service Commands - Ubuntu / WSL2

```
sudo service mysql start
sudo service mysql status
```

- `start` -- begin running the server (does nothing if already running).
- `status` -- check whether it's currently running.

```
sudo service mysql stop
sudo service mysql restart
```

- `stop` -- shut it down. Use only when you intentionally want the database unavailable.
- `restart` -- stop then start again. Use after a configuration change, or when troubleshooting.

MySQL Service Commands - macOS

```
brew services start mysql
brew services list
```

- `start` -- begin running the server as a background service (does nothing if already running).
- `list` -- Homebrew's `services` has no single-service `status` command; `list` shows the status of every

```
brew services stop mysql
brew services restart mysql
```

- `stop` -- shut it down. Use only when you intentionally want the database unavailable.
- `restart` -- stop then start again. Use after a configuration change, or when troubleshooting.

Using the MySQL Shell

In **Verify on Ubuntu / WSL2**, `-e` runs one SQL command directly from the terminal, then returns to the regular prompt:

```
sudo mysql -e "SELECT VERSION();"
```

Use the MySQL shell when you want to run several SQL commands in one interactive session.

Ubuntu / WSL2

```
sudo mysql
```

Ubuntu/WSL2 normally uses `auth_socket` for the MySQL `root` account, so `sudo mysql` confirms your Linux administrator access.

macOS

```
mysql -u root
```

Homebrew MySQL on macOS normally allows `mysql -u root` without a password. If you set a root password on macOS, use `mysql -u root -p` instead.

RECOMMENDED

Recommended: Use a Separate App Account

For this course, do not set a MySQL `root` password. Use `root` only to administer MySQL; do not use it in a PHP application.

RECOMMENDED

Database Account

After opening the MySQL shell as `root`, create an account with access only to its course database:

- PHP should connect as `course_app`, using its password, rather than as `root`.

```
CREATE DATABASE course_db;
CREATE USER 'course_app'@'localhost' IDENTIFIED BY 'your-strong-password';
GRANT SELECT, INSERT, UPDATE, DELETE ON course_db.* TO 'course_app'@'localhost';
```

Inside the MySQL Shell

The terminal prompt changes to `mysql>`. Try these commands:

```
SHOW DATABASES;
SELECT VERSION();
EXIT;
```

`EXIT;` returns you to the regular terminal.

OPTIONAL

Uninstall MySQL on Ubuntu / WSL2 (Optional)

Ubuntu / WSL2: Stop the server, then remove MySQL and its configuration files.

```
sudo service mysql stop
sudo apt remove --purge -y mysql-server mysql-client mysql-common
sudo apt autoremove -y
```

OPTIONAL

Uninstall MySQL on macOS (Optional)

macOS: Stop the Homebrew service, then remove MySQL.

```
brew services stop mysql
brew uninstall mysql
```